JAMA Network Open

Original Investigation | Psychiatry

Problematic Social Media Use and Attention-Deficit/Hyperactivity Disorder Symptoms in Adolescents

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Abstract

IMPORTANCE Given shared processes in attentional control, impulsivity, and reward sensitivity, problematic social media use (PSMU) may be associated with attention-deficit/hyperactivity disorder (ADHD) symptoms. Yet most evidence has been from cross-sectional studies; multiyear prospective, within-person studies on an association of PSMU with ADHD symptoms are lacking.

OBJECTIVES To estimate bidirectional, within-person associations between PSMU and ADHD symptoms across 5 annual assessments, and to test variation by sex.

DESIGN, SETTING, AND PARTICIPANTS This prospective cohort study used data from 5 annual assessments (year 2 [2017-2019] through year 6 [2021-2023]) of the Adolescent Brain Cognitive Development Study, a population-based cohort that enrolled youths aged 9 to 10 years from 21 US sites in year 1 (2016-2018). Participants had data on ADHD symptoms or PSMU at 1 or more waves between year 2 and year 6. Data analysis was performed between July and September 2025.

EXPOSURE Youth-reported PSMU measured using the Social Media Addiction Questionnaire.

MAIN OUTCOMES AND MEASURES Parent-reported ADHD symptoms (measured using the ADHD Problems subscale of the Child Behavior Checklist) at each wave. Random-intercept cross-lagged panel models were used to separate stable between-person differences from within-person variation and to estimate concurrent and lagged within-person associations between PSMU and ADHD symptoms across waves.

RESULTS The analytic sample consisted of 11 286 youths with a mean (SD) age at year 2 of 12.1 (0.67) years; 5918 (52.4%) were male. At the within-person level, years in which youths reported higher-than-usual PSMU were followed by elevations in parent-reported ADHD symptoms (year 4 to year 5: β = 0.08 [95% CI, 0.03-0.12], $P < .001$; year 5 to year 6: β = 0.08 [95% CI, 0.03-0.13], $P = .001$). ADHD symptoms were also associated with higher PSMU from year 4 to year 5 (β = 0.04 [95% CI, 0.01-0.07]; $P = .02$). Tests of moderation indicated that the prospective association between PSMU and ADHD symptoms was higher in magnitude for males than for females (change in $\chi^2_{16} = 31.7$; $P = .01$).

CONCLUSIONS AND RELEVANCE This cohort study of US youths found that year-to-year elevations in self-reported PSMU were directly associated with ADHD symptoms during mid-adolescence, particularly among males; reverse associations of ADHD symptoms with later PSMU were small in magnitude and sporadic. Screening for hard-to-control social media use, time spent on social media, and attention to sex-specific patterns may be warranted, although the population-based associations may be small.

Key Points

Question Are within-person changes in problematic social media use (PSMU) among youths associated with attention-deficit/hyperactivity disorder (ADHD) symptoms across 5 annual study waves, and do these associations differ by sex?

Findings In this cohort study of 11 286 youth participants, within-person elevations in self-reported PSMU behaviors were associated with higher parent-reported ADHD symptoms the following year during mid-adolescence. These within-person associations differed by sex and were significant among males but not females.

Meaning The findings suggest that assessment of social media use among youths with ADHD should prioritize problematic patterns and consider sex-specific risk.

+ Supplemental content

Author affiliations and article information are listed at the end of this article.

JAMA Network Open. 2026;9(7):e2623748. doi:10.1001/jamanetworkopen.2026.23748

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Introduction

Attention-deficit/hyperactivity disorder (ADHD), characterized by persistent and impairing patterns of reduced concentration and/or impulsivity,[1] remains one of the most common neurodevelopmental disorders of youth. Approximately 7 million youths in the US aged 3 to 17 years (11.4%) have ever received an ADHD diagnosis, representing an increase of 1 million cases from 2016 to 2022.[2] Among these cases, nearly 8 in 10 presented with at least 1 co-occurring condition, such as behavioral disorders, anxiety, or depression.[2-4] During this period, youths have also experienced broader social, economic, and technological changes, including the continued growth of digital media use, particularly social media. These concurrent patterns support the need for longitudinal examinations of the extent to which social media behavior and ADHD symptoms exhibit time-ordered coupling within individuals across adolescence.

Large cohort studies have reported that engagement with multiple digital media activities is associated with later ADHD symptoms,[5,6] and other studies have reported that social media use is associated with attention problems and broader mental health difficulties.[7,8] More recent longitudinal studies have adopted designs that separate stable between-person differences from within-person change and, in some cases, have evaluated neuropsychological mechanisms that could support an association between media exposure and ADHD.[7-9] For example, using multilevel models to decompose between-person and within-person components of screen time, Wallace et al[9] found that social media, television, and video games were directly associated with ADHD symptoms in a prospective cohort of Canadian high school students; impulsivity and, to a lesser extent, response inhibition and working memory accounted for a substantial portion of these associations.[9]

Exposure-based measures (eg, hours per day on social media or other screen activities) do not directly capture the extent to which youths experience their social media use as distressing or interfering with daily functioning.[10-16] Problematic social media use (PSMU), sometimes referred to as social media addiction, is characterized by loss of control over social media behavior resulting in an unfavorable pattern of engagement (eg, preoccupation and unsuccessful attempts to cut down) and adverse consequences.[10,12-14] Conceptually, these features align closely with ADHD-related impairments in self-regulation, impulsivity, and heightened sensitivity to immediate rewards.[17-20] Research also interprets impulsivity and neurocognitive control problems as key liabilities underlying associations between heavier social media exposure and ADHD symptoms[9,21]; the construct of PSMU reflects those motivational and control processes directly (eg, unsuccessful attempts to cut down), rather than treating exposure time as a proxy for them.[12-16] Conceptually, PSMU should be related to ADHD symptoms at the trait level and may show different longitudinal patterns because it more directly reflects how self-regulatory and reward systems are engaged by social media use.[10,11,22]

Features of adolescence provide a context in which ADHD-related vulnerabilities and PSMU can be mutually reinforcing. Adolescence is a developmental period during which intensifying responsiveness to rewards outpaces maturation of self-regulation systems.[7,23,24] Moreover, youths spend considerable time on social media platforms, which deliver rapid, socially contingent feedback and continuous opportunities for engagement. These affordances align with ADHD-related tendencies toward heightened reward sensitivity and difficulties delaying gratification and are implemented through social media platform features, such as likes, notifications, and infinite scrolling.[25,26] At times when ADHD symptoms are particularly pronounced for youths, momentary regulatory control may be further compromised,[17-20] making disengagement from intermittent social rewards and algorithmically personalized content especially difficult.[27,28] This inability to disengage may trigger compulsive or habitual checking and preoccupation with social media.[29] The reverse pathway is also plausible: greater PSMU may be associated with concurrent task load (eg, multitasking across platforms) and exposure to more emotionally arousing content.[8,30,31] These processes likely tax working memory and shorten sustained attention, placing additional demands on regulatory systems already characterized by deficits in attentional control and impulse regulation.

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Prior longitudinal work[5,32] has typically characterized digital media behavior using exposure-based indicators (eg, screen time and number of activities). Moreover, studies of PSMU have mostly focused on associations with depressive symptoms, suicidal ideation, or self-injurious behavior and have rarely incorporated multiyear, within-person models or ADHD outcomes.[12-16,33] In these latter studies, problematic or addictive patterns of technology use tend to show more robust and more consistent associations with adverse outcomes than do simple exposure measures, suggesting that indices of PSMU provide information beyond time-based indicators. However, it remains unclear as to what extent PSMU and ADHD symptoms exhibit time-ordered coupling within individuals across adolescence, which requires moving beyond cross-sectional and purely between-person analyses.

Methodological work has emphasized the importance of separating trait-like differences from state-like fluctuations in longitudinal data,[34-36] and recent studies of digital media and ADHD have begun to adopt such approaches.[9] In line with these recommendations, the present investigation aimed to estimate bidirectional, within-person associations between PSMU and ADHD symptoms across 5 annual assessments and to test variation by sex. Specifically, we applied random-intercept cross-lagged panel models (RI-CLPMs) to partition stable between-person differences in ADHD symptoms and PSMU from within-person deviations over time and to test whether years in which youths report higher-than-usual PSMU are followed by changes in ADHD symptoms (and vice versa) after accounting for stable vulnerability.[37]

We also considered possible sex-based differences, as ADHD is more frequently diagnosed in males than in females.[38,39] Males tend to show higher levels of hyperactive-impulsive and externalizing behaviors than females,[40,41] and experimental and neuroimaging work suggests sex-related differences in reward-related circuitry, delay discounting, and sensitivity to immediate rewards among youths with ADHD.[42-44] Yet limited prior work has specifically examined sex-based differences in the coupling between ADHD symptoms and PSMU; thus, our sex-stratified models are best viewed as exploratory (rather than confirmatory) tests.[11,22,45,46]

Two competing hypotheses follow from the literature. First, PSMU is prospectively associated with higher subsequent ADHD symptoms, such that periods of more dysregulated social media engagement may precede worsening attentional difficulties. Second, ADHD symptoms are prospectively associated with higher subsequent PSMU, such that impulsivity and reward sensitivity may precede more problematic social media engagement. We expected both within-person pathways to be present but small in magnitude, with the PSMU-to-ADHD direction showing somewhat greater magnitude and more consistent prospective associations.

Methods

Study Participants and Analytic Sample

We analyzed data from the July 2025, 6.0 Release of the Adolescent Brain Cognitive Development (ABCD) Study, a US population–based, multisite cohort that enrolled approximately 11 868 youths aged 9 to 10 years.[47] The University of California, San Diego Central Institutional Review Board approved the ABCD Study protocol. Parents or caregivers provided written informed consent, and youths assented.[48] In accordance with 45 CFR § 46.104, this cohort study was exempt from ethics review and informed consent requirement because it used deidentified data. We followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline.

Participants in the ABCD Study were enrolled from 2016 to 2018 (year 1) from 21 sites and were assessed annually thereafter; the present study analyzed data spanning year 2 (2017-2019) through year 6 (2021-2023). Only partial year 6 data had been released at the time of analysis, accounting for the larger reduction in available participants at that wave compared with years 2 through 5. Analyses included youths with data on ADHD symptoms or PSMU at 1 or more waves between year 2 and year 6. Full-information maximum likelihood estimation allowed inclusion of participants with partial follow-up data.[49]

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Measures

Youth participants completed the Social Media Addiction Questionnaire, a 6-item self-report scale adapted from the Bergen Facebook Addiction Scale.[50] The questionnaire is rated on a 1 (indicating never) to 6 (indicating very often) scale and averaged to a total score, and items capture social media use–related preoccupation, tolerance, escape, difficulty cutting back, distress from inability to use, and perceived adverse consequences; higher scores indicate greater PSMU (eMethods 2-6 in Supplement 1). The questionnaire was administered only to youths reporting at least 1 social media account at a given wave; those without an account did not complete the questionnaire, and their PSMU score at that wave was treated as structurally missing. Measurement invariance of the PSMU scale across sex was supported at the metric and scalar levels in multigroup confirmatory factor analyses (eMethods 6, eTable 4, and eFigure 1 in Supplement 1).

ADHD symptoms were reported by parents or caregivers via the *Diagnostic and Statistical Manual of Mental Disorders*–oriented ADHD Problems subscale of the Child Behavior Checklist. A dichotomous ADHD diagnosis and medication status at year 2 were defined using the 5-item Kiddie Schedule for Affective Disorders and Schizophrenia (KSADS-5) ADHD module and ABCD Study prescription records, respectively. Details on ADHD diagnosis and medication coding are provided in the eMethods 5 in Supplement 1.

Covariates

Models were adjusted for sex, race and ethnicity (dummy-coded with 6 race categories—American Indian or Alaska Native, Asian, Black or African American, Native Hawaiian or Other Pacific Islander, non-Hispanic White [reference group], and multiracial), baseline annual household income, and parental educational level. Race and ethnicity were included because ADHD diagnosis, access to treatment, and social media use patterns vary across racial and ethnic groups in the US and may confound associations between PSMU and ADHD symptoms. Sensitivity analyses excluded youths with a KSADS-5 ADHD diagnosis or ADHD medication use at year 2. Covariate coding details are provided in eMethods 8 in Supplement 1, and the sensitivity sample exclusion criteria are provided in eMethods 12 (eTable 6) in Supplement 1.

Statistical Analysis

Longitudinal associations between PSMU and ADHD symptoms across 5 annual waves were examined using CLPMs and RI-CLPMs in R version 4.5.1 (R Core Team),[51] with the lavaan package version 0.6-19.[52] We used robust maximum likelihood estimation[53] and accommodated missing data using full-information maximum likelihood under a missing-at-random assumption.[54] Youths with intermittent participation or changes in social media account status contributed all observed ADHD and PSMU data, and waves in which youths had no accounts were treated as structurally missing for PSMU (eMethods 1 in Supplement 1). To examine sex-based differences, we estimated multigroup RI-CLPMs with sex as the grouping factor.[55] Additional details on missing data and sample selection (including a cohort flow diagram) are provided in eMethods 9 and eFigure 4 in Supplement 1, and structural model specifications are provided in eMethods 10 and eTable 5 in Supplement 1.

Pearson correlations (r) were calculated to describe bivariate associations among observed ADHD symptom and PSMU scores. RI correlations and within-wave residual correlations were estimated as parameters in the RI-CLPMs.

For interpretive context, standardized longitudinal cross-lagged coefficients of $\beta = 0.03$, $\beta = 0.07$, and $\beta = 0.12$ were interpreted as small, medium, and large effect sizes, respectively (eMethods 11 in Supplement 1).[56] In a sensitivity analysis, we repeated the RI-CLPM using youth-reported ADHD symptoms from the Youth Self-Report at years 4 through 6 (eMethods 13 in Supplement 1). The testing was 2-sided, and $P < .05$ was considered statistically significant. Given the dependence structure among structural equation modeling parameters, we did not apply corrections for multiple comparisons because these can be overly conservative in this context.[57] Instead, interpretation focused on a limited set of theory-driven cross-lagged coefficients and their

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consistency across model specifications. Data analysis was performed between July and September 2025.

Results

The analytic cohort had 11 286 participants (mean [SD] age, 12.1 [0.67] years at year 2), of whom 5368 (47.8%) were female and 5918 were male (52.4%) at baseline (**Table 1**). Descriptive statistics for PSMU and ADHD symptoms, and their bivariate correlations, are reported in eTables 1-3 in Supplement 1. Across waves, PSMU scores remained low, with annual mean (range) of 1.77 (1.47-1.96) on the 1 to 6 scale. Moreover, the score thresholds corresponding to the 85th and 95th percentiles showed little change from year 2 to year 6, indicating that the distribution of PSMU did not shift meaningfully upward across adolescence (eMethods 7, eFigure 2, and eFigure 3 in Supplement 1). The main model incorporated 44 067 total observations across the 5 annual waves (mean [range] observations per participant, 3.9 [1-5]), with concurrent ADHD and PSMU data available for 10 865 at year 2, 10 180 at year 3, 9448 at year 4, 8619 at year 5, and 4955 participants at year 6.

Model selection favored the RI-CLPM framework. In fit comparisons spanning years 2 to 6, the unconstrained RI-CLPM showed excellent absolute fit (χ^2_{91} = 691.010, P < .001; comparative fit index [CFI] = 0.986; Tucker-Lewis index [TLI] = 0.979; root mean square error of approximation [RMSEA] = 0.024 (90% CI, 0.023-0.026); standardized root mean square residual [SRMR] = 0.022), and it fit significantly better than the constrained RI-CLPM with time-invariant lagged paths (change in χ^2_{12} = 142.59; P < .001) (eTable 5 in Supplement 1). Conventional CLPMs fit worse overall (eg, constrained CLPM: χ^2_{68} = 4001.99 [P < .001], CFI = 0.91, RMSEA = 0.072 [90% CI, 0.070-0.073];

Table 1. Sociodemographic Characteristics of the Analytic Sample at Year 2

Characteristic	Participants, No. (%) (N = 11 286)[a]
Age, mean (SD), y	12.1 (0.67)
Sex	
Female	5368 (47.8)
Male	5918 (52.4)
Race[b]	
American Indian or Alaska Native	57 (0.5)
Asian	243 (2.2)
Black or African American	1771 (15.7)
Native Hawaiian or Other Pacific Islander	15 (0.1)
White	7515 (66.6)
Multiracial	1232 (10.9)
Other[c]	451 (4.1)
Ethnicity[b]	
Hispanic	2289 (20.3)
Non-Hispanic	8963 (79.4)
Annual household income at baseline, $	
<25 000	1485 (13.2)
25 000-49 999	1507 (13.4)
50 000-74 999	1422 (12.6)
75 000-99 999	1524 (13.5)
100 000-199 999	3222 (28.5)
≥200 000	1207 (10.7)
Parental educational level	
≤High school diploma	1569 (13.9)
≥College degree	9702 (86.0)

[a] Numbers may not sum to the total due to missing data.

[b] Race and ethnicity were self-reported by participants.

[c] Includes other, unknown, refused to answer, or not reported.

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unconstrained CLPM: χ^2_{56} = 3893.86 [*P* < .001], CFI = 0.91, RMSEA = 0.078 [90% CI, 0.076-0.080]); thus, primary associations focused on the unconstrained RI-CLPM results. In a sensitivity analysis excluding youths with a parent-reported ADHD diagnosis or ADHD medication use at year 2, model fit and estimates were similar (eTable 6 in Supplement 1); unless otherwise noted, the results presented here were derived from the inclusive models.

Within-person stability increased for both PSMU and ADHD symptoms. Autoregressive carryover for PSMU strengthened over time (β = 0.24 [95% CI, 0.21-0.28] for year 2 to year 3, 0.28 [95% CI, 0.25-0.31] for year 3 to year 4, 0.41 [95% CI, 0.38-0.44] for year 4 to year 5, and 0.43 [95% CI, 0.39-0.46] for year 5 to year 6; all *P* < .001). ADHD symptoms showed reliable autoregressive carryover across waves, with associations ranging from β = 0.15 (95% CI, 0.08-0.21; *P* < .001) to 0.35 (95% CI, 0.31-0.38 (*P* < .001) (**Table 2**, **Figure 1**). Within-person coupling was concentrated in later waves and was greater in magnitude from PSMU to ADHD symptoms than in the reverse direction. Increases in PSMU at 1 wave were associated with higher ADHD symptoms at the next wave from year 4 to year 5 (β = 0.08 [95% CI, 0.03-0.12]; *P* < .001) and from year 5 to year 6 (β = 0.08 [95% CI, 0.03-0.13]; *P* = .001). The reciprocal path from ADHD symptoms to subsequent PSMU was significant only from year 4 to year 5 (β = 0.04 [95% CI, 0.01-0.07]; *P* = .02) and was otherwise small and not statistically different from 0 (Table 2). Overall, cross-lagged coefficients were small in magnitude, and most other lags were near 0, suggesting a time-limited coupling rather than a pervasive longitudinal association.

Significant between-person and concurrent associations were also present. There was a positive RI correlation between trait ADHD symptoms and trait PSMU (*r* = 0.08 [95% CI, 0.04-0.11]; *P* < .001) (Table 2, Figure 1). Same-wave residual covariances were positive from year 2 to year 5 (*r* = 0.04 [95% CI, 0.01-0.07; *P* = .02] to 0.08 [95% CI, 0.04-0.11; *P* < .001]) but attenuated by year 6 (*r* = 0.01 [95% CI, −0.03 to 0.05]; *P* = .63), indicating concurrent cofluctuation earlier in adolescence that was no longer evident by the final wave.

Sex-stratified RI-CLPMs demonstrated good fit for the configural (freely estimated) model (CFI = 0.988, TLI = 0.973; RMSEA = 0.056 [90% CI, 0.050-0.062]; SRMR = 0.028) (**Table 3**). Equality tests favored a model that allowed lagged paths to vary by sex (change in χ^2_{16} = 31.7; *P* = .01), indicating that at least some autoregressive or cross-lagged coefficients differed by sex. Differences were concentrated in the PSMU to ADHD symptoms direction (**Figure 2**, Table 3). Among males, increases in PSMU were associated with higher subsequent ADHD symptoms at 2 consecutive lags, from year 4 to year 5 (β = 0.11 [95% CI, 0.02-0.19]; *P* = .01) and from year 5 to year 6 (β = 0.07 [95% CI, 0.00-0.14]; *P* = .04). No PSMU to ADHD symptoms paths were significant among females (Table 3), and ADHD symptoms to PSMU cross-lagged associations were not significant in either sex. The correlation between trait ADHD symptoms and trait PSMU was significant among females (β = 0.12 [95% CI, 0.00-0.23]; *P* = .04) but not among males (β = 0.07 [95% CI, −0.01 to 0.14]; *P* = .10).

In sensitivity RI-CLPM analyses using self-reported ADHD symptoms from the Youth Self-Report at year 4 to year 6, we again observed a trait-level association between ADHD symptoms and PSMU (eTable 7, eTable 8, and eFigure 5 in Supplement 1). However, the within-person pattern differed by informant (youth or parent). In the parent-reported models, higher-than-usual PSMU had small direct associations with parent-reported ADHD symptoms at the next assessment (eg, year 4 to year 5: β = 0.08 [95% CI, 0.03-0.12]; *P* < .001). In contrast, in the youth-reported models, year-to-year changes in PSMU were not meaningfully associated with self-reported ADHD ratings; instead, small associations were observed from higher ADHD symptoms to later PSMU. For example, youth-reported ADHD symptoms at year 4 were associated with PSMU at year 5 (β = 0.098; B [SE] = 0.23 [0.07]; *P* < .001). In other words, parents tended to report slightly higher ADHD-like behavior following periods when youths reported elevated PSMU, whereas youths did not report corresponding changes in their own symptoms over the same intervals.

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Table 2. Standardized Estimates From RI-CLPMs of Attention-Deficit/Hyperactivity Disorder Symptoms and Problematic Social Media Use[a]

Variable	β coefficient (95% CI)	B (SE)	*P* value
Race and ethnicity[b]			
American Indian or Alaska Native			
RI of ADHD symptoms	0.02 (−0.01 to 0.04)	0.49 (0.41)	.24
RI of PSMU	0.03 (0.00 to 0.06)	0.22 (0.11)	.04
Asian			
RI of ADHD symptoms	−0.05 (−0.07 to −0.04)	−0.77 (0.12)	<.001
RI of PSMU	0.006 (−0.02 to 0.03)	0.02 (0.04)	.56
Black or African American			
RI of ADHD symptoms	−0.05 (−0.07 to −0.02)	−0.28 (0.07)	<.001
RI of PSMU	0.19 (0.16 to 0.22)	0.25 (0.02)	<.001
Native Hawaiian or Other Pacific Islander			
RI of ADHD symptoms	−0.01 (−0.03 to 0.01)	−0.84 (0.57)	.14
RI of PSMU	−0.01 (−0.03 to 0.00)	−0.16 (0.11)	.13
Non-Hispanic White			
RI of ADHD symptoms	0.04 (0.02 to 0.06)	0.22 (0.06)	.001
RI of PSMU	−0.07 (−0.10 to −0.04)	−0.09 (0.02)	<.001
Multiracial			
RI of ADHD symptoms	0.04 (0.02 to 0.07)	0.31 (0.08)	<.001
RI of PSMU	0.06 (0.03 to 0.08)	0.09 (0.02)	<.001
Other race			
RI of ADHD symptoms	−0.02 (−0.04 to −0.00)	−0.23 (0.12)	.049
RI of PSMU	0.03 (−0.00 to 0.06)	0.07 (0.04)	.07
Annual household income[c]			
RI of ADHD symptoms	−0.15 (−0.17 to −0.12)	−0.19 (0.02)	<.001
RI of PSMU	−0.11 (−0.14 to −0.07)	−0.03 (0.005)	<.001
Parental educational level[d]			
RI of ADHD symptoms	0.08 (0.05 to 0.10)	0.49 (0.07)	<.001
RI of PSMU	−0.06 (−0.09 to −0.03)	−0.08 (0.02)	<.001
Autoregressive path[e]			
ADHD symptoms year 2 → year 3	0.35 (0.31 to 0.38)	0.33 (0.02)	<.001
ADHD symptoms year 3 → year 4	0.21 (0.16 to 0.26)	0.18 (0.02)	<.001
ADHD symptoms year 4 → year 5	0.15 (0.08 to 0.21)	0.14 (0.03)	<.001
ADHD symptoms year 5 → year 6	0.20 (0.13 to 0.27)	0.20 (0.04)	<.001
PSMU year 2 → year 3	0.24 (0.21 to 0.28)	0.31 (0.03)	<.001
PSMU year 3 → year 4	0.28 (0.25 to 0.31)	0.28 (0.02)	<.001
PSMU year 4 → year 5	0.41 (0.38 to 0.44)	0.44 (0.02)	<.001
PSMU year 5 → year 6	0.43 (0.39 to 0.46)	0.40 (0.02)	<.001
Cross-lagged path[e]			
ADHD symptoms year 2 → PSMU year 3	−0.002 (−0.03 to 0.03)	−0.001 (0.007)	.92
ADHD symptoms year 3 → PSMU year 4	0.004 (−0.03 to 0.04)	0.002 (0.009)	.82
ADHD symptoms year 4 → PSMU year 5	0.04 (0.01 to 0.07)	0.02 (0.01)	.02
ADHD symptoms year 5 → PSMU year 6	0.01 (−0.03 to 0.05)	0.007 (0.01)	.63
PSMU year 2 → ADHD symptoms year 3	0.02 (−0.01 to 0.05)	0.05 (0.04)	.23
PSMU year 3 → ADHD symptoms year 4	0.03 (−0.01 to 0.06)	0.05 (0.03)	.11
PSMU year 4 → ADHD symptoms year 5	0.08 (0.03 to 0.12)	0.12 (0.03)	<.001
PSMU year 5 → ADHD symptoms year 6	0.08 (0.03 to 0.13)	0.12 (0.04)	.001

(continued)

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Table 2. Standardized Estimates From RI-CLPMs of Attention-Deficit/Hyperactivity Disorder Symptoms and Problematic Social Media Use[a] (continued)

Variable	β coefficient (95% CI)	B (SE)	*P* value
Within-wave covariance path[e]			
RI of ADHD symptoms ↔ RI of PSMU	0.08 (0.04 to 0.11)	0.08 (0.02)	<.001
ADHD symptoms year 2 ↔ PSMU year 2	0.04 (0.01 to 0.07)	0.04 (0.02)	.02
ADHD symptoms year 3 ↔ PSMU year 3	0.07 (0.04 to 0.10)	0.09 (0.02)	<.001
ADHD symptoms year 4 ↔ PSMU year 4	0.06 (0.03 to 0.10)	0.07 (0.02)	<.001
ADHD symptoms year 5 ↔ PSMU year 5	0.08 (0.04 to 0.11)	0.08 (0.02)	<.001
ADHD symptoms year 6 ↔ PSMU year 6	0.01 (−0.03 to 0.05)	0.01 (0.02)	.63

Abbreviations: ADHD, attention-deficit/hyperactivity disorder; PSMU, problematic social media use; RI, random intercept; RI-CLPM, random-intercept cross-lagged panel model.

[a] Unconstrained RI-CLPM fit: χ^2_{91} = 691.010, *P* < .001; comparative fit index = 0.986; Tucker-Lewis index = 0.979; root mean square error of approximation = 0.024 (90% CI, 0.023-0.026); standardized root mean square residual = 0.022. 95% CIs are for unstandardized estimates.

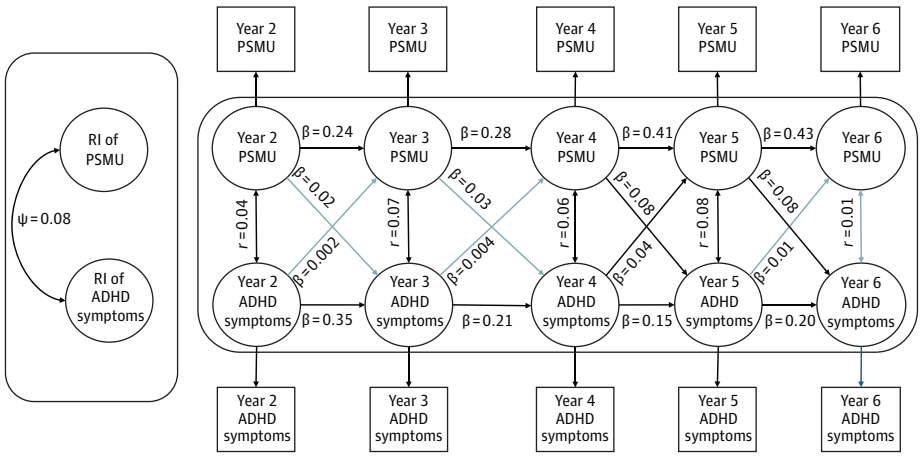
[b] Race was entered as a dummy variable. Ethnicity was coded 0 for Hispanic and 1 for non-Hispanic.

[c] Annual household income was coded 1 to 6 (1 for<$25 000; 6 for≥$200 000) and treated as ordinal.

[d] Parental educational level was coded 0 for high school diploma or less and 1 for college degree or more.

[e] Right arrow indicates a prospective regression path from one annual assessment to the next. A bidirectional arrow indicates a within-wave covariance or RI covariance.

Figure 1. Path Diagram of Within-Person Associations Between Attention-Deficit/Hyperactivity Disorder (ADHD) Symptoms and Problematic Social Media Use (PSMU) From Years 2 to 6



Random-intercept cross-lagged panel model (RI-CLPM) estimates included 11 286 participants of the ABCD Study. Circles represent within-person deviations in ADHD symptoms and PSMU at each year, and rectangles represent observed scores. Horizontal arrows are standardized autoregressive paths (β), diagonal arrows are standardized cross-lagged paths (β), and vertical arrows are within-wave correlations (*r*); blue arrows are not statistically significant (*P* ≥ .05). The left side of the diagram shows RIs capturing stable between-person differences, with ψ indicating their correlation. Race and ethnicity, annual household income, and parental educational level were included as time-invariant covariates at the between-person level. Unconstrained RI-CLPM fit: χ^2_{91} = 691.010, *P* < .001; comparative fit index = 0.986; Tucker-Lewis index = 0.979; root mean square error of approximation = 0.024 (90% CI, 0.023-0.026); standardized root mean square residual = 0.022.

Discussion

Partially supporting our hypothesis, the present study found standardized coefficients in the small to medium range, time-limited longitudinal associations between PSMU and ADHD symptoms in youth participants of the ABCD Study. Within-person analyses showed that elevations in PSMU were associated with slightly higher ADHD symptom severity at the next annual assessment during mid-adolescence (approximately ages 14 to 16 years), after accounting for stable trait differences and prior symptom levels. These cross-lagged associations were in the small-to-medium range by RI-CLPM benchmarks,[46,56,58] which is expected given that the model removes stable between-person variance and the cross-informant design reduces shared method variance. However, the clinical significance at the individual level remains uncertain. Many other cross-lagged paths, especially in early adolescence and in the ADHD to PSMU direction, were close to 0 and

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Table 3. Sex-Stratified Autoregressive and Cross-Lagged Associations Between Attention-Deficit/Hyperactivity Disorder Symptoms and Problematic Social Media Use[a]

Path and group[b,c]	β coefficient (95% CI)	B (SE)	*P* value
Autoregressive[d]			
ADHD symptoms year 2 → year 3			
Female	0.30 (0.24 to 0.36)	0.29 (0.03)	<.001
Male	0.35 (0.30 to 0.40)	0.33 (0.03)	<.001
ADHD symptoms year 3 → year 4			
Female	0.20 (0.12 to 0.27)	0.18 (0.04)	<.001
Male	0.20 (0.13 to 0.28)	0.17 (0.03)	<.001
ADHD symptoms year 4 → year 5			
Female	0.19 (0.10 to 0.28)	0.20 (0.05)	<.001
Male	0.13 (0.04 to 0.22)	0.11 (0.04)	.005
ADHD symptoms year 5 → year 6			
Female	0.30 (0.21 to 0.39)	0.29 (0.05)	<.001
Male	0.14 (0.04 to 0.24)	0.15 (0.05)	.006
PSMU year 2 → year 3			
Female	0.31 (0.26 to 0.36)	0.38 (0.03)	<.001
Male	0.24 (0.18 to 0.29)	0.29 (0.04)	<.001
PSMU year 3 → year 4			
Female	0.34 (0.30 to 0.39)	0.35 (0.02)	<.001
Male	0.21 (0.16 to 0.26)	0.21 (0.03)	<.001
PSMU year 4 → year 5			
Female	0.44 (0.40 to 0.48)	0.47 (0.02)	<.001
Male	0.33 (0.28 to 0.37)	0.36 (0.03)	<.001
PSMU year 5 → year 6			
Female	0.44 (0.39 to 0.49)	0.41 (0.03)	<.001
Male	0.37 (0.31 to 0.43)	0.36 (0.03)	<.001
Cross-lagged[d]			
ADHD symptoms year 2 → PSMU year 3			
Female	−0.001 (−0.04 to 0.04)	−0.001 (0.01)	.95
Male	0.01 (−0.03 to 0.05)	0.005 (0.008)	.59
ADHD symptoms year 3 → PSMU year 4			
Female	−0.003 (−0.07 to 0.06)	−0.002 (0.02)	.92
Male	0.05 (−0.01 to 0.11)	0.02 (0.01)	.08
ADHD symptoms year 4 → PSMU year 5			
Female	0.06 (−0.00 to 0.11)	0.04 (0.02)	.07
Male	0.04 (−0.02 to 0.09)	0.02 (0.02)	.18
ADHD symptoms year 5 → PSMU year 6			
Female	0.002 (−0.07 to 0.07)	0.001 (0.02)	.96
Male	0.02 (−0.06 to 0.09)	0.009 (0.02)	.68
PSMU year 2 → ADHD symptoms year 3			
Female	0.02 (−0.02 to 0.07)	0.05 (0.05)	.29
Male	0.03 (−0.01 to 0.07)	0.09 (0.06)	.18
PSMU year 3 → ADHD symptoms year 4			
Female	0.04 (−0.03 to 0.10)	0.06 (0.05)	.26
Male	0.04 (−0.01 to 0.08)	0.07 (0.05)	.16
PSMU year 4 → ADHD symptoms year 5			
Female	0.06 (−0.03 to 0.14)	0.09 (0.07)	.18
Male	0.11 (0.02 to 0.19)	0.18 (0.08)	.01
PSMU year 5 → ADHD symptoms year 6			
Female	0.07 (−0.00 to 0.15)	0.11 (0.06)	.05
Male	0.07 (0.00 to 0.14)	0.12 (0.06)	.04

(continued)

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Table 3. Sex-Stratified Autoregressive and Cross-Lagged Associations Between Attention-Deficit/Hyperactivity Disorder Symptoms and Problematic Social Media Use[a] (continued)

Path and group[b,c]	β coefficient (95% CI)	B (SE)	*P* value
Within-wave covariance[d]			
RI of ADHD symptoms ↔ RI of PSMU			
Female	0.12 (0.00 to 0.23)	0.12 (0.06)	.04
Male	0.07 (−0.01 to 0.14)	0.07 (0.04)	.10
ADHD symptoms year 2 ↔ PSMU year 2			
Female	0.04 (−0.05 to 0.12)	0.04 (0.05)	.37
Male	0.03 (−0.03 to 0.09)	0.03 (0.03)	.38
ADHD symptoms year 3 ↔ PSMU year 3			
Female	0.09 (0.04 to 0.14)	0.10 (0.03)	.001
Male	0.08 (0.04 to 0.12)	0.09 (0.03)	<.001
ADHD symptoms year 4 ↔ PSMU year 4			
Female	0.05 (−0.03 to 0.12)	0.05 (0.04)	.22
Male	0.10 (0.04 to 0.16)	0.10 (0.03)	.002
ADHD symptoms year 5 ↔ PSMU year 5			
Female	0.04 (−0.02 to 0.09)	0.04 (0.03)	.17
Male	0.11 (0.06 to 0.17)	0.11 (0.03)	<.001
ADHD symptoms year 6 ↔ PSMU year 6			
Female	−0.002 (−0.06 to 0.06)	−0.002 (0.03)	.95
Male	0.01 (−0.05 to 0.07)	0.01 (0.03)	.72

Abbreviations: ADHD, attention-deficit/hyperactivity disorder; PSMU, problematic social media use; RI, random intercept.

[a] Multigroup random-intercept cross-lagged panel model (configural/free) fit: scaled χ^2_{38} = 276.900; comparative fit index = 0.988; Tucker-Lewis index = 0.973; root mean square error of approximation = 0.056 (90% CI, 0.050-0.062); standardized root mean square residual = 0.028. Change in χ^2_{16} = 31.700 (*P* = .01). 95% CIs for parameter estimates are unstandardized.

[b] Female n = 5368; male n = 5918 (used cases).

[c] Arrows indicate the direction of the year-to-year path from the variable and year on the left to the variable and year on the right.

[d] Right arrow indicates a prospective regression path from one annual assessment to the next. A bidirectional arrow indicates a within-wave covariance or RI covariance.

nonsignificant, indicating that any within-person coupling is concentrated in specific mid-adolescent intervals.

There was comparatively limited support for ADHD symptoms being associated with later PSMU, although 1 cross-lagged path was significant from year 4 to year 5 (approximately ages 14 to 15 years) but was no longer evident from year 5 to year 6. The significant between-person association indicates that youths with chronic ADHD symptoms generally exhibit greater PSMU. In contrast, the within-person findings suggest that periods of youth-reported higher-than-usual PSMU tend to be followed by slightly higher parent-reported ADHD symptoms at the next wave. Because these estimates come from observational data, we interpreted them as time-ordered associations rather than causal associations, and we cannot exclude time-varying confounding.

In sensitivity RI-CLPMs using youth-reported ADHD symptoms, we again found a moderate trait-level association between ADHD symptoms and PSMU, but the within-person pattern differed. Parent-reported models showed PSMU and ADHD symptoms associations, whereas youth self-reported models showed little evidence that year-to-year shifts in PSMU were followed by changes in their own ADHD ratings, with small associations instead shown from ADHD symptoms to later PSMU. This pattern might suggest that parents are somewhat more sensitive to changes in outward behavior and regulation around social media use, whereas youths' self-reports may reflect ongoing attentional and impulsivity difficulties that precede, rather than follow, more problematic engagement. This informant-dependent pattern indicates that the association between ADHD symptoms and PSMU is likely more nuanced than any single model can capture and that a more comprehensive investigation into mechanisms, context, and functional outcomes is needed.

Developmental Timing and Sex-Based Differences

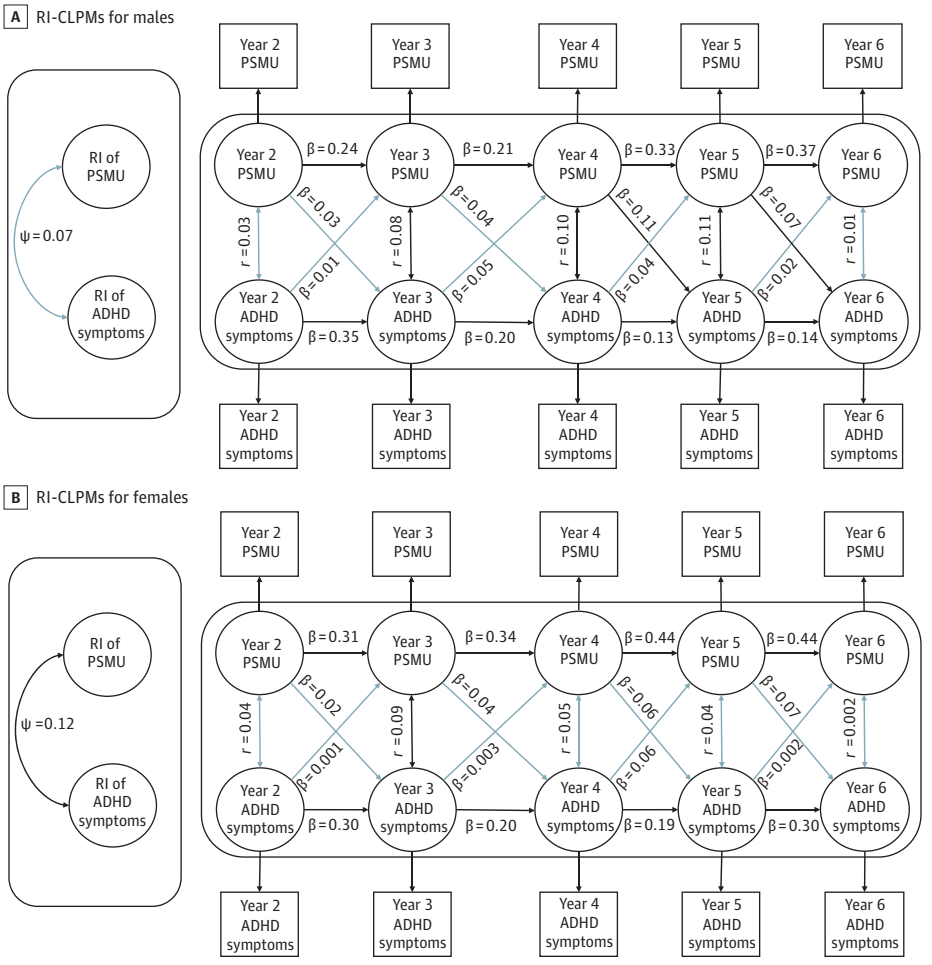
An important contribution of this study to the literature is it identified that detectable within-person coupling between PSMU and ADHD symptoms is small and concentrated in mid-adolescence. This nonstationarity suggests that, as youths age and gain more autonomy over social media use and time online more generally,[59] the association between PSMU and ADHD symptoms becomes evident. By mid-adolescence, a majority of youths have social media accounts,[60,61] and many are online

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frequently; for those with underlying attention or impulse-control vulnerabilities, mid-adolescence may be a period when PSMU and ADHD-related difficulties are more likely to move together over time. This pattern aligns with developmental neuroscience, with adolescence characterized by heightened reward reactivity and sensation-seeking alongside still-maturing cognitive control systems.[62] This dual systems imbalance is at its most sensitive in the mid-adolescence years.[11,63]

In sex-stratified models, the prospective association of PSMU with later ADHD symptoms was significant only for males and remained small in magnitude and limited to mid-adolescent lags. Adolescent males may experience a somewhat tighter link between more PSMU and subsequent attention problems compared with females, at least within this age range. However, we found a significant positive correlation between trait ADHD and trait PSMU in females, suggesting that females with greater chronic ADHD symptoms exhibit more PSMU than those with fewer ADHD symptoms. This pattern aligns with prior work showing inconsistent patterns regarding sex as a moderator of associations between ADHD symptoms and digital media, with greater apparent vulnerability for males in some contexts[64] (eg, violent media) and for females in other contexts[65,66] (eg, media multitasking). Given that standardized coefficients were small in magnitude and the limited prior theory on sex-specific ADHD and PSMU pathways, these sex-based differences should be viewed as exploratory and need replication and further investigation.[67-70]

Figure 2. Path Diagram of Sex-Stratified Within-Person Association Between Attention-Deficit/Hyperactivity Disorder (ADHD) Symptoms and Problematic Social Media Use (PSMU) From Years 2 to 6



Random-intercept cross-lagged panel model (RI-CLPM) estimates were separate for 5918 male (A) and 5368 female (B) participants of the ABCD Study. Circles represent within-person deviations in ADHD symptoms and PSMU at each year, and rectangles represent observed scores. Horizontal arrows are standardized autoregressive paths (β), diagonal arrows are standardized cross-lagged paths (β), and vertical arrows are within-wave correlations (*r*); blue arrows are not statistically significant ($P \geq .05$). The left panel shows RIs capturing stable between-person differences, with ψ indicating their correlation. Race and ethnicity, annual household income, and parental educational level were included as time-invariant covariates at the between-person level in all models. Multigroup random-intercept cross-lagged panel model (configural/free) fit: scaled $\chi^2_{38} = 276.900$; comparative fit index = 0.988; Tucker-Lewis index = 0.973; root mean square error of approximation = 0.056 (90% CI, 0.050-0.062); standardized root mean square residual = 0.028.

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Limitations

While this study leveraged a large sample and rigorous modeling, several limitations warrant discussion. First, ADHD symptoms were parent-reported, which tends to capture more impairment than youth self-report and is often more closely align with clinician assessments.[71] Thus, observed associations may partly reflect reporter perception rather than behavioral change. Second, race and ethnicity also were captured via self-reported categories, which may not fully reflect heterogeneity, and the ABCD Study sample overrepresents White youths from higher-income families; thus, findings may not be as generalizable to racial and ethnic minority youths or youths from lower-income families. Third, the PSMU measure also aggregates across platforms (eg, TikTok, Snapchat, YouTube, and multiplayer gaming chat applications) that differ in behavioral affordances and user demographic characteristics; differences in platform preference between males and females may partially account for sex-stratified associations. Fourth, we did not empirically test the incremental validity of PSMU over simple social media exposure time, and future work should directly compare these measures within the same model. Annual measurement intervals are also too coarse to capture transient within-person associations between social media engagement and attention or impulsivity that operate on hourly or daily timescales. Future studies could incorporate more frequent and/or objective measures of social media use (eg, screenomics),[72] such as smartphone use logs or ecological momentary assessment of in-the-moment attention lapses, to triangulate these findings.[73,74]

Fifth, the observational design of the study precluded causal conclusions, as time-varying confounders can alter associations.[75,76] Although RI-CLPM controlled for stable between-person differences and we addressed many time-invariant confounders via RIs, time-varying confounders were not modeled and may account for some or all of the observed within-person associations. Sixth, although there are several plausible pathways linking PSMU and ADHD symptoms (ie, disrupted attention control due to multitasking, heightened reward craving, or sleep disruption from late-night use),[8,77,78] we did not directly measure these potential mediators.

Conclusions

In this large cohort study of US youths followed across 5 annual study waves, within-person elevations in PSMU were associated with small subsequent increases in parent-reported ADHD symptoms during mid-adolescence, with associations most evident among male participants. Reverse associations of ADHD symptoms with later PSMU were small in magnitude and inconsistent. These findings suggest that PSMU may warrant attention in clinical assessments of youths, particularly those with attentional challenges, although the small in magnitude and time-limited nature of the within-person associations indicate that broader clinical or policy implications should be considered cautiously.

ARTICLE INFORMATION

Accepted for Publication: May 19, 2026.

Published: July 17, 2026. doi:10.1001/jamanetworkopen.2026.23748

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Obtained funding: Nagata, Baker.

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Conflict of Interest Disclosures: Dr Nagata reported receiving grants from the National Institutes of Health (NIH) during the conduct of the study. Dr Christakis reported serving as an expert witness in ongoing social media litigation outside the submitted work. Dr Baker reported receiving grants from the NIH during the conduct of the study. No other disclosures were reported.

Funding/Support: This research was supported by grant K08HL159350 from the NIH (Dr Nagata).

Role of the Funder/Sponsor: The funder had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Data Sharing Statement: See Supplement 2.

Disclaimer: The opinions and assertions expressed herein are those of the authors and do not reflect the official policy or position of the Uniformed Services University of the Health Sciences or the US Department of Defense.

Additional Information: The ABCD Study was supported by the NIH and additional federal partners under award numbers U01DA041022, U01DA041025, U01DA041028, U01DA041048, U01DA041089, U01DA041093, U01DA041106, U01DA041117, U01DA041120, U01DA041134, U01DA041148, U01DA041156, U01DA041174, U24DA041123, and U24DA041147. A full list of supporters is available at https://abcdstudy.org/federal-partners/. A listing of participating sites and a complete listing of the study investigators can be found at https://abcdstudy.org/principal-investigators.html. ABCD consortium investigators designed and implemented the study and/or provided data but did not necessarily participate in analysis or writing of this report.

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SUPPLEMENT 1.

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SUPPLEMENT 2.

Data Sharing Statement

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